

Grade 1 Ontario Curriculum Spatial Sense

How Long? How Big?

Unit Overview: How Long? How Big?

Grade: Grade 1 | **Strand:** Spatial Sense

Target Expectations: E2.1, E2.2, E2.3

Learning Goal: Children will identify measurable attributes, compare and order objects by attributes, and read a calendar.

Lesson	Lesson Title	Learning Goal
Day 1	Name the Attribute	I can name measurable attributes of objects.
Day 2	Compare by Length and Area	I can compare and order objects by length and area.
Day 3	Compare by Mass and Capacity	I can compare and order objects by mass and capacity.
Day 4	Read the Calendar	I can read the date and find days, weeks, months on a calendar.
Day 5	Measurement Day Capstone	I can use measurement and the calendar together.

Lesson 1: Name the Attribute

Goal: Every object has MEASURABLE ATTRIBUTES: length, area, mass, capacity.

Expectation: E2.1

Learning Goal: "I can name measurable attributes of objects."

Materials: M1_attribute_cards, M7_attribute_chart, M10_unit_poster

1. Minds On (12 mins): Roo Names the Attributes

Roo: This pencil has LENGTH. This book has AREA. This rock has MASS. This cup has CAPACITY! Roo opens: 'Measure it!' Class: 'Match it!' Cal: 'Mark the day!'

Bring children to the carpet. Roo and Cal puppets. 'Friends — I'm Roo the Ruler Rabbit. Today we name MEASURABLE ATTRIBUTES of objects.'

2. Action (25 mins): Attribute Stations

- Whole group: name attributes for 6 classroom objects.
- Three stations.
- Stretch challenge at Station C: name 5 attributes for one object (length, area, mass, capacity, plus colour or shape).
- Discuss attribute choice.
- Circulate to record diagnostic evidence on the attribute tracker.
- STRATEGY COMPARE at Station C: two pairs share how they decided which attribute to name for an AMBIGUOUS object (backpack — mass? capacity? both?).

3. Consolidation (13 mins): Roo's Attribute Brag

Discuss:

- What is an ATTRIBUTE?
- What are the 4 measurable attributes?
- Can one object have many attributes?
- When do we use CAPACITY?
- Roo's preview: tomorrow we'll COMPARE lengths and areas (Lesson 2 — E2.1).

Exit: Roo's unit refrain — Roo calls 'Measure it!' Class chants 'Match it!' Cal echoes 'Mark the day!' Repeat 3x.

Lesson 2: Compare by Length and Area

Goal: We line up to compare LENGTH. We cover to compare AREA.

Expectation: E2.2

Learning Goal: "I can compare and order objects by length and area."

Materials: M2_length_strips, M3_area_tiles, M7_attribute_chart, M10_unit_poster

1. Minds On (12 mins): Roo Lines Up the Strips

Roo: I have 4 ribbons. Line them up — which is LONGEST? Now look at 4 cards — which has the BIGGEST AREA?

Bring children to the carpet. Roo and Cal puppets. 'Friends — yesterday we named attributes. Today we COMPARE LENGTH and AREA.'

2. Action (25 mins): Length and Area Stations

- Whole group: order 4 ribbons by length, 4 cards by area.
- Three stations.
- Stretch challenge at Station C: order 6 objects.
- Discuss the line-up and cover rules.
- Circulate to record formative evidence on the tracker.

3. Consolidation (13 mins): Roo's Compare Brag

Discuss:

- How do we compare LENGTH?
- How do we compare AREA?
- What does ORDER mean?
- What was easiest?

Exit: Each child writes ONE comparison: 'My ___ is longer than my ___.'

Lesson 3: Compare by Mass and Capacity

Goal: We use a balance for MASS. We pour for CAPACITY.

Expectation: E2.2

Learning Goal: "I can compare and order objects by mass and capacity."

Materials: M4_balance_mat, M5_cup_cards, M7_attribute_chart, M10_unit_poster

1. Minds On (12 mins): Roo Balances and Pours

Roo: I'll BALANCE these 4 objects to find the heaviest. Then I'll POUR to find the cup that holds the most!

Bring children to the carpet. Roo and Cal puppets. 'Friends — yesterday length and area. Today MASS and CAPACITY.'

2. Action (25 mins): Mass and Capacity Stations

- Whole group: order 4 objects by mass, 4 cups by capacity.
- Three stations.
- Stretch challenge at Station C: order 6 objects.
- Discuss balance and pour rules.
- Circulate to record formative evidence.

3. Consolidation (13 mins): Roo's Mass and Capacity Brag

Discuss:

- How do we compare MASS?
- How do we compare CAPACITY?
- What does it mean to ORDER 4?
- Which was easier — mass or capacity?

Exit: Each child writes ONE comparison: 'My ___ is heavier than my ___.'

Lesson 4: Read the Calendar

Goal: A CALENDAR shows DAYS, WEEKS, MONTHS, HOLIDAYS, and SEASONS.

Expectation: E2.3

Learning Goal: "I can read the date and find days, weeks, months on a calendar."

Materials: M6_calendar_mat, M8_calendar_chart, M10_unit_poster

1. Minds On (12 mins): Cal Reads the Calendar

Cal: TODAY is the 4th of May! May is a MONTH. There are 7 DAYS in a WEEK. Let's find some HOLIDAYS!

Bring children to the carpet. Roo and Cal puppets. 'Friends — yesterday mass and capacity. Today CALENDAR.'

2. Action (25 mins): Calendar Stations

- Whole group: find today, this week, this month.
- Three stations.
- Stretch challenge at Station C: name the SEASON for each holiday.
- Discuss the calendar grid.
- Circulate to record formative evidence.

3. Consolidation (13 mins): Cal's Calendar Brag

Discuss:

- What is a CALENDAR?
- How many days in a week?
- Name a Canadian holiday and its season.
- What season are we in?

Exit: Each child writes today's date: 'Today is ___ (day) ___ (date) ___ (month).'

Lesson 5: Measurement Day Capstone

Goal: We name attributes, compare and order objects, and use a calendar to mark our measurement day.

Expectation: E2.1, E2.2, E2.3

Learning Goal: "I can use measurement and the calendar together."

Materials: M1_attribute_cards, M2_length_strips, M3_area_tiles, M4_balance_mat

1. Minds On (12 mins): Roo and Cal Plan Measurement Day

Roo: Today is MEASUREMENT DAY! 4 stations: NAME the attribute, ORDER by length, ORDER by mass, READ the calendar! Roo opens: 'Measure it!' Class: 'Match it!' Cal: 'Mark the day!'

Bring children to the carpet. Roo and Cal puppets in lab coats. 'Friends — today is our CAPSTONE. We use ALL we know: attributes, comparisons, calendar.'

2. Action (25 mins): Measurement Day

- WHOLE GROUP: Roo + Cal review Day 1-4 success criteria.
- Station 1 — NAME THE ATTRIBUTE: 8 mixed objects.
- Station 2 — ORDER BY LENGTH (transitivity): pair has 4 objects.
- Station 3 — ORDER BY MASS (sense-making): pair has 4 objects of similar size but different mass (water bottle, brick fragment, sponge, pencil case).
- Station 4 — MARK THE DAY: Cal's calendar.
- Circulate to record formative evidence on the capstone rubric.

3. Consolidation (13 mins): Roo and Cal's Capstone Brag

Discuss:

- Which station was your favourite?
- What's the difference between mass and capacity?
- What did you learn about the calendar?
- How would you measure a desk?

Exit: Roo + Cal closing refrain — Roo: 'Measure it!' Class: 'Match it!' Cal: 'Mark the day!' Repeat 3x.

Name: _____

Date: _____

Roo's Attribute Hunt

Learning Goal: I can name measurable attributes of objects.

Mission

1. Draw a line from each object to its main attribute (length / area / mass / capacity).
2. Pick ONE object.
3. Find 3 things at home.

Part 1: Match the attribute.

Draw a line from each object to its main attribute (length / area / mass / capacity).

[illegible]

Name: _____

Date: _____

Roo's Length and Area Sheet

Learning Goal: I can compare and order objects by length and area.

Instructions for the Student:

1. Number the strips 1 (shortest) to 4 (longest).
2. Count the tiles covering each shape.
3. Find 3 things at home.

Part 2: Count tiles.

Count the tiles covering each shape. Write the area in tiles.

[illegible]

Date: _____

Learning Goal: I can compare and order objects by mass and capacity.

1. Number 4 objects from lightest (1) to heaviest (4).
2. Number 4 cups from smallest (1) to largest (4) capacity.
3. Pick 3 things in your kitchen.

Pick 3 things in your kitchen. Order them by mass OR capacity.

[illegible]

Date: _____

Cal's Calendar Reader

Learning Goal: I can read the date and find days, weeks, months on a calendar.

Mission

1. Look at the calendar.
2. Count days in the month.
3. Name 2 Canadian holidays.

Part 1: Find dates.

Look at the calendar. Mark **TODAY** with a star. Mark next **FRIDAY** with a heart.

[illegible]

Name: _____

Date: _____

Roo and Cal's Measurement Day

Learning Goal: I can use measurement and the calendar together.

Instructions for the Student:

1. Pick 2 objects.
2. Order 4 strips by length (1-4).
3. Find today on the calendar.

Part 3: CALENDAR.

Find today on the calendar. Mark it. Name a holiday and its season.

Attribute Cards

(M1_attribute_cards)

A set of 16 cards with attributes (length, area, mass, capacity) and example objects.

Benchmarks

1 cm fingernail	10 cm thumb to pinky
1 m long step	1 g paper clip
100 g stick of butter	1 kg 1 L of water
1 mL drop	1 L milk carton

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 10 sets (1 per pair).

2. Laminate before cutting.

3. Cut into 16 cards per set.

4. Store each set in a labelled bag.

USE IN CLASS

Quantity: 10 sets

Lessons: 1, 5

Prep: ~15 min

Length Strips

(M2_length_strips)

A set of 24 paper strips of varied lengths for length comparison.

Centimetre Strips



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

TEACHER PREP

1. Print 10 sets.

2. Laminate before cutting.

3. Cut into 24 strips per set.

4. Store in labelled bag.

USE IN CLASS

Quantity: 10 sets

Lessons: 2, 5

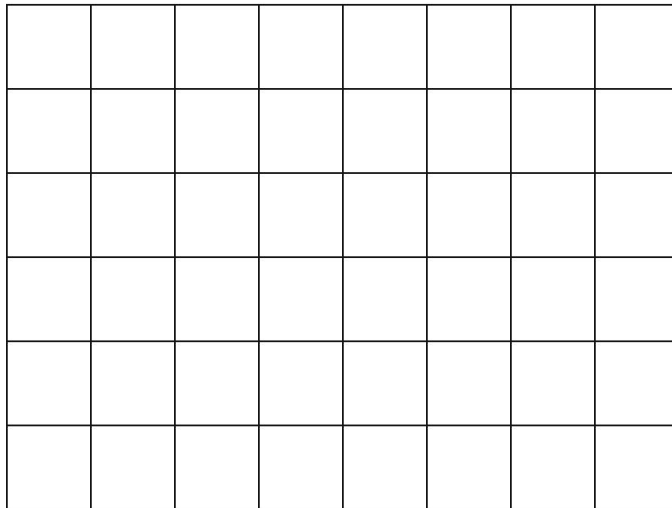
Prep: ~25 min

Area Tiles

(M3_area_tiles)

A set of 60 paper unit tiles for covering areas.

Area Grid — 8 × 6



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

TEACHER PREP

1. Print 10 sets.

2. Laminate.

3. Cut into 60 tiles per set.

4. Store in labelled bag.

USE IN CLASS

Quantity: 10 sets

Lessons: 2, 5

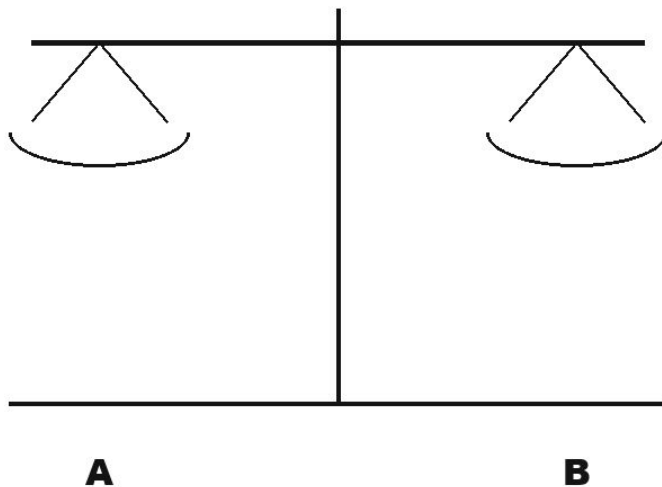
Prep: ~30 min

Paper Balance Mat

(M4_balance_mat)

A letter-size mat with a balance for mass comparison.

Balance Scale



PRINT SPECS

Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

TEACHER PREP

1. Print 12 mats.
2. Laminate.
3. Distribute one per pair.

USE IN CLASS

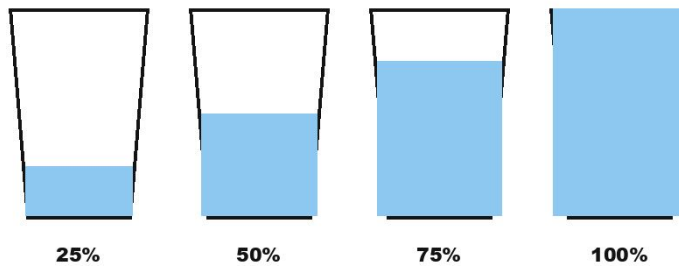
Quantity: 10-12 mats
Lessons: 3, 5
Prep: ~15 min

Capacity Cup Cards

(M5_cup_cards)

A set of 12 paper cup cards in 3 sizes for capacity activities.

Cups & Capacity



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 10 sets.

2. Laminate.

3. Cut into 12 cups per set.

4. Store in labelled bag.

USE IN CLASS

Quantity: 10 sets

Lessons: 3, 5

Prep: ~20 min

Calendar Mat

(M6_calendar_mat)

A letter-size monthly calendar template.

Calendar — April

Sun	Mon	Tue	Wed	Thu	Fri	Sat
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 12 mats.

2. Laminate for dry-erase.

3. Distribute one per pair.

USE IN CLASS

Quantity: 10-12 mats

Lessons: 4, 5

Prep: ~15 min

Measurable Attributes Chart

(M7_attribute_chart)

Anchor chart of 5 attributes.

Benchmarks

1 cm fingernail	10 cm thumb to pinky
1 m long step	1 g paper clip
100 g stick of butter	1 kg 1 L of water
1 mL drop	1 L milk carton

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Laminate.

3. Tape to front wall before Lesson 1.

USE IN CLASS

Quantity: 1 chart

Lessons: 1, 2, 3, 5

Prep: ~10 min

Calendar Reference Chart

(M8_calendar_chart)

Anchor chart with days, weeks, months, seasons, and Canadian holidays.

Calendar — April

Sun	Mon	Tue	Wed	Thu	Fri	Sat
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Laminate.

3. Tape to front wall.

USE IN CLASS

Quantity: 1 chart

Lessons: 4, 5

Prep: ~10 min

Measurement Day Template

(M9_capstone_template)

Letter-size capstone template with 4 station response areas.

Capstone

Capstone canvas — measure 5 things.

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: no

TEACHER PREP

1. Print 25 templates.

2. Stack near teacher table.

3. No lamination.

USE IN CLASS

Quantity: 25 templates

Lessons: 5

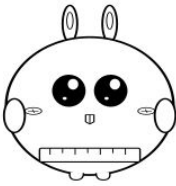
Prep: ~5 min

How Long? How Big? Poster

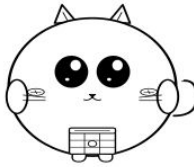
(M10_unit_poster)

Unit anchor poster of Roo and Cal.

How Long? How Big? Poster



Roo the Ruler Rabbit



Cal the Calendar Cat

Colourful Ruler

Wall Calendar

Length

Area

PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Laminate.

3. Tape to front wall on Day 1.

USE IN CLASS

Quantity: 1 poster

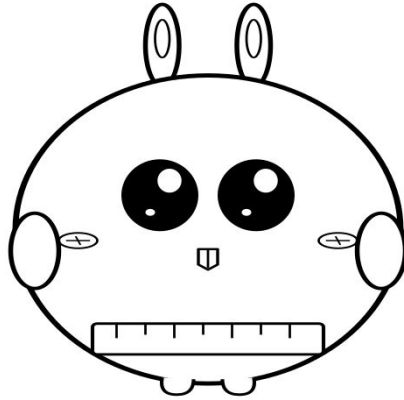
Lessons: 1, 2, 3, 4, 5

Prep: ~10 min

Roo the Ruler Rabbit Puppet

(char_roo_puppet)

Paper puppet of Roo for teacher use in all 5 lessons.



Roo the Ruler Rabbit

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Cut around silhouette + tab.

3. Laminate.

4. Glue popsicle stick to tab.

USE IN CLASS

Quantity: 1 puppet

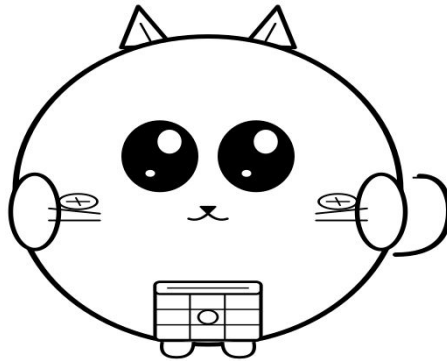
Lessons: 1, 2, 3, 4, 5

Prep: ~12 min

Cal the Calendar Cat Puppet

(char_cal_puppet)

Paper puppet of Cal for teacher use in Lessons 4-5.



Cal the Calendar Cat

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 copy.

2. Cut around silhouette + tab.

3. Laminate.

4. Glue popsicle stick to tab.

USE IN CLASS

Quantity: 1 puppet

Lessons: 4, 5

Prep: ~12 min

Name: _____

Date: _____

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
E2.1 identify measurable attributes of two-dimensional shapes and three-dimensional objects, including length, area, mass, capacity, and angle	Names 0-1 attributes consistently. Confuses mass and capacity.	Names 2-3 attributes with prompts.	Independently names all 4 attributes (length, area, mass, capacity) and picks the right one for an object.	Names multiple attributes per object AND mentions angle as a 5th attribute.
E2.2 compare several everyday objects and order them according to length, area, mass, and capacity	Compares pairs but cannot order 4+ objects. Often relies on visual judgment.	Orders 3 objects with prompts. Uses correct technique for one attribute.	Independently orders 4+ objects across all 4 attributes using correct techniques (line up, cover, balance, pour).	Orders 6+ objects accurately AND verifies with a second technique.
E2.3 read the date on a calendar, and use a calendar to identify days, weeks, months, holidays, and seasons	Cannot consistently read today's date or find days/weeks/months.	Reads today's date with prompts. Names a season or holiday.	Independently reads dates, navigates forward and backward, names a holiday and its season.	Identifies dates 2 weeks ahead and matches Canadian holidays to seasons.

Junior Measure-Master Certificate

This certificate is proudly presented to:

For successfully completing the 5-Day How Long? How Big? unit and demonstrating the skills of a Junior Measure-Master. A measurement certificate. Roo the Ruler Rabbit congratulates the student.

By completing this unit, this student has demonstrated:

- Naming measurable attributes
- Ordering objects by length and area
- Ordering objects by mass and capacity
- Reading and using a calendar

Marketplace Listing

A 5-day Grade 1 unit on non-standard measurement framed around Roo the Ruler Rabbit and Cal the Calendar Cat.

Price: \$8.99 CAD

Pages: ~42

Classroom time: 250 min

Teacher prep: ~180 min

Top tags: Grade 1, math, measurement, non-standard, length, area, mass, capacity

What's included:

- Unit blueprint with curriculum mapping (1 file)
- 5 lesson plans (5 files)
- 5 student worksheets (5 files)
- Manipulatives plan with 12 print-ready assets (1 file)
- Formative + reflection sheets (1 file)
- Assessment suite (1 file)